Modeling Destruction: An Empirical Analysis on the 2002 Gujarat Riots

ECON 580

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Motivation and Question

Question: **What impact does social and physical composition of a community have on intensity of inter-communal riots following an exogenous catalyzing incident in Indian state of Gujarat?**

Motivation

- 2002 Riots indicator of inter-communal relations in 21st century
- Estimated 60 million USD in damages, 1000+ people killed
- Riots deter investment from firms

Results Preview

Limited economic significance and no statistical significance across all factors I investigate for inter-communal violence in Gujarat

History of Inter-Communal Violence

Gujarat Inter-Communal Violence

- Before independence, both Hindu and Muslim Kingdoms
- Partition causes migration of Muslims
- Violent rioting starts in 1969

2002 Godhra Overview

- Feburary 27, 2002 Train of pilgrims arriving from Ayodhya
- Train attacked by mob of 2000+ people
- Caused largest riots in Gujarat since 1969

2002 Riots

Theory on causes and effects of 2002 riots

- Muslim religious monuments desecrated – Dargah, Madrasa, Masjid
- Gujarati media non-objective on riots against Muslims (Bormann, Cedarman, Vogt 2015)
- Internal migration of Muslims in Gujarat (Chandhoke 2009)
- Government used riots as harmful intervention in Muslim communities

Figure: Districts in Gujarat experiencing riots

Data Overview

Datasets

- 2001 Census
  - Mother-Tongue Language
  - Population of District
  - Proportion of Hindu and Muslim Population
- 1991 Census
  - Change in Muslim Population
- Citizens for Justice and Peace (CJP)
  - Religious Monuments Destroyed/Desecrated

Cleaning Issues

- Different Number of Districts from 1991 and 2001
- Missing Data for Sabarkantha District
- Manually insert data for 8 districts

Empirical Model

Regression Specification

$$
\begin{aligned}
MuslimDestruction_i = \beta_0 &+ \beta_1\, pctGujarati_i + \beta_2\, pctMuslimChange_i \\
&+ \beta_3 (pctHinduMinusMuslim_i)^2 \\
&+ \beta_4\, goodHousing_i \\
&+ \epsilon_i
\end{aligned}
$$

Variables

$MuslimDestruction_i$	Muslim sites destroyed per 100k population in district i
$pctGujarati_i$	Share of Gujarati mother-tongue speakers, 2001
$pctMuslimChange_i$	Change in urban Muslim population share, 1991–2001
$(pctHinduMinusMuslim_i)^2$	Squared Hindu–Muslim population share difference
$goodHousing_i$	Good-condition housing units per capita

Regression Results: Baseline Models

	(1)	(2)	(3)	(4)
Intercept	-3.012	4.507^{**}	1.544	9.821^{*}
	(8.020)	(1.347)	(7.869)	(3.590)
% Gujarati	0.085		0.104	
	(0.093)		(0.088)	
Δ Muslim %		0.079		0.045
		(0.180)		(0.175)
(Hindu $-$ Muslim %)^2			$-0.001^{\cdot}$	-0.001
			(0.001)	(0.001)
R^2	0.036	0.009	0.189	0.120
Adj. R^2	-0.008	-0.038	0.112	0.032
N	24	23	24	23

$^{\cdot}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Outcome: Muslim sites destroyed per 100,000 population

Regression Results: Full Models

	(5)	(6)	(7)
Intercept	0.856	6.475	0.656
	(7.793)	(4.169)	(8.700)
% Gujarati	0.076		0.079
	(0.090)		(0.104)
Δ Muslim %		0.012	0.052
		(0.172)	(0.181)
(Hindu − Muslim %)^2	-0.001^*	-0.001	-0.001
	(0.001)	(0.001)	(0.001)
Good Housing per Capita	6.960	8.995	6.988
	(5.633)	(6.129)	(6.730)
R^2	0.246	0.210	0.235
Adj. R^2	0.133	0.085	0.064
N	24	23	23

$^*p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$, $^{****}p < 0.001$

Outcome: Muslim sites destroyed per 100,000 population

Limitations

- Large outlier
  - Anand District
- Association vs. Causation
  - Higher percentage of Muslims likely means more Muslim Religious Monuments
- Economic Significance vs. Statistical Significance
  - Signs follow theory
  - Nothing statistically significant
- Small data sample
  - Want Taluka level (sub district)

Anand District Outlier

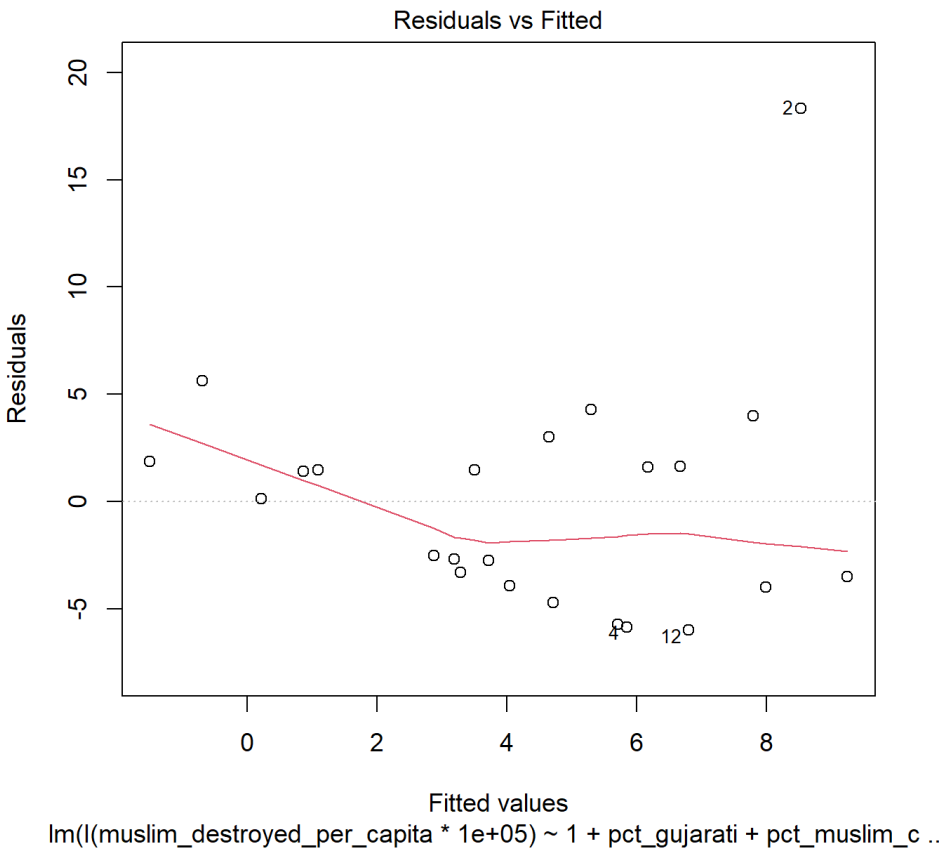

Figure: Residuals vs Fitted – Anand District = 2

Implications

- Somewhat follows theory
  - Positive relationship for Gujarati speakers and migration
  - Small negative relationship for different of Hindu and Muslim
- More empirical work needs to be done
  - Media distribution
  - Migration of Muslims
  - Further exploration
- Theory on inter-communal violence from **specifically** 2002 Gujarat should be revisited

Conclusion

- Limited economic significance and no statistical significance
- Signs follow theory
- Improvements for final paper
  - Explain Anand Outlier
  - Include maps on 1991-2001 migration
  - Find data for Sabarkantha District